Christu Jyothi Institute of Technology & Science

JANGAON

COMPUTER SCIECNE ENGINEERING DEPARTMENT

B.TECH – III YEAR II SEMESTER

(Regulation: R22)

(2025-2026)

FULL STACK DEVELOPMENT LAB MANUAL

(CS611PE)

List of Experiments

1. Create an application to setup node JS environment and display “Hello World”.

2. Create a Node JS application for user login system.

3. Write a Node JS program to perform read, write and other operations on a file.

4. Write a Node JS program to read form data from query string and generate response using NodeJS

5. Create a food delivery website where users can order food from a particular restaurant listed in the website for handling http requests and responses using NodeJS.

6. Implement a program with basic commands on databases and collections using MongoDB.

7. Implement CRUD operations on the given dataset using MongoDB.

8. Perform Count, Limit, Sort, and Skip operations on the given collections using MongoDB.

9. Develop an angular JS form to apply CSS and Events.

10. Develop a Job Registration form and validate it using angular JS.

11. Write an angular JS application to access JSON file data of an employee from a server using $http service.

12. Develop a web application to manage student information using Express and Angular JS.

13. Write a program to create a simple calculator Application using React JS.

14. Write a program to create a voting application using React JS

15. Develop a leave management system for an organization where users can apply different types of leaves such as casual leave and medical leave. They also can view the available number of days using react application.

16. Build a music store application using react components and provide routing among the pages.

17. Create a react application for an online store which consist of registration, login, product information pages and implement routing to navigate through these pages.

EXPERIMENT – 1

1. Create an application to setup node JS environment and display “Hello World”.

AIM: To set up the Node.js development environment and create a simple application that displays **“Hello World”** on the console and/or browser.

OBJECTIVE:

- To install and configure Node.js.
- To initialize a Node.js project.
- To write a basic JavaScript program using Node.js.
- To execute a Node program in the terminal.
- To create a simple HTTP server (optional).

THEORY:

Node.js is an open-source, cross-platform runtime environment built on Chrome’s V8 JavaScript engine. It allows developers to execute JavaScript code outside the browser. Node.js is widely used for backend development, API creation, and real-time applications.

Key points:

- Uses asynchronous & event-driven architecture
- High performance due to V8 engine
- Supports large ecosystem (NPM modules)

PROCEDURE

Step 1: Install Node.js (first you need to install Python in the system).

Type the command at command prompt: **C:\> pip install nodejs-bin (After Install check below commands)**

C:\> node –v
C:> npm -v

Step 2: Create Project Folder

C:\> md exp1
C:\> cd exp1

Step 3: Initialize Node Project

C:/exp1>npm init -y

Step 4: Create `App.js` File

Write the following code: (Notepad or visual Studio IDE)

```
const http = require("http");

const server = http.createServer((req, res) =>
   { res.writeHead(200, { "Content-Type": "text/plain" });
   res.end("Hello World");
});

server.listen(3000, () => {
   console.log("Server running at http://localhost:3000");
});
```

Step 5: Run the Application (Goto the command prompt at same directory)

Type following command:

C:\exp1> node App.js
Server running at http://localhost:3000
(Copy given URL at Browser Address Bar for Output).

```
suresh@Suresh:~/Desktop/FSD/programs/exp1$ node App.js
Server running at http://localhost:3000
```

Hello World

Viva Question and Answers :

1. **What is Node.js?**
   Node.js is a JavaScript runtime environment.
   It runs JavaScript outside the browser.
2. **What is V8 engine?**
   V8 is Google's JavaScript engine.
   It converts JS into machine code.
3. **What is npm?**
   npm stands for Node Package Manager.
   It manages Node.js packages.
4. **What is `require()`?**
   It imports modules in Node.js.
   Example: `require("http")`.
5. **What is `http.createServer()`?**
   It creates a web server.
   It handles request and response.
6. **What is `res.writeHead()`?**
   It sets status code and headers.
   Example: 200 OK.
7. **What is `res.end()`?**
   It sends response to client.
   It ends the request cycle.
8. **What is port number?**
   Port identifies a service on server.
   Example: 3000.
9. **What is asynchronous programming?**
   Non-blocking execution model.
   Improves performance.
10. **How to run Node program?**
    Use command `node filename.js`.
    It executes JavaScript file.

EXPERIMENT – 2

2. Create a Node JS application for user login system.

AIM:

To create a simple Node.js application that connects to MySQL to register a user (with hashed password) and allow login verification.

OBJECTIVE

- Connect Node.js app with MySQL database.
- Create `users` table.

THEORY (brief)

Node.js allows creating servers using the built-in **`http`** module. The **MySQL2** module enables Node.js to connect to a MySQL database.

PROCEDURE:

1. Create project folder and initialize:

```
C:\> md exp2 C:\
>cd exp2 C:\
exp2>npm init -y
```

Process:

1. Browser sends login form data (POST request).
2. Node.js server reads the request body manually using events (`data` and `end`).
3. The `querystring` module converts the POST data into key-value format.
4. A SQL query checks whether username & password exist in the MySQL `users` table.
5. The server responds with either *Login Successful* or *Invalid Credentials*.

This experiment shows how backend validation works without frameworks.

PREPARATION / SQL (run in MySQL):
Open Xampp Application, Start Service Apache, Mysql Services..
Click In Mysql service Open admin(http://localhost/phpmyadmin).

```
CREATE DATABASE IF NOT EXISTS exp2;
USE exp2;

CREATE TABLE IF NOT EXISTS users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(100) NOT NULL UNIQUE,
  email VARCHAR(255) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO `users` (`id`, `username`, `email`, `password`, `created_at`) VALUES (NULL,
'kamal', 'kamal@gmail.com', '1234', current_timestamp());
```

PROCEDURE

1. Install Node.js and MySQL on the system.
2. Create project folder and files:
   - `server.js`
   - `login.html`
   - `styles.css`
3. Install MySQL module:

   C:\>md exp2

   c:\>cd exp2

   C:\exp2> npm install mysql2

4. Start MySQL and create table using SQL above.
5. Write code in `server.js` to:

   - Connect to database
   - Serve HTML and CSS files
   - Receive POST request
   - Validate credentials

6. Run server:

   C:\exp2>node server.js

7. Open browser:

   http://localhost:3000

PROGRAM(CODE):

Server.js:

```
const http = require("http");
const fs = require("fs");
const querystring = require("querystring");
const mysql = require("mysql2");

// MySQL connection
const db =
   mysql.createConnection({ host:
   "localhost",
   user: "root",
   password: "",
   database: "exp2"
});

db.connect((err) =>
   { if (err) throw err;
   console.log("Connected to MySQL");
});

const server = http.createServer((req, res) => {

   if (req.method === "GET" && req.url === "/")
      { fs.readFile("login.html", (err, data) => {
         res.writeHead(200, { "Content-Type": "text/html" });
         res.end(data);
      });
```

```
    }

    else if (req.method === "GET" && req.url === "/styles.css")
        { fs.readFile("styles.css", (err, data) => {
            res.writeHead(200, { "Content-Type": "text/css" });
            res.end(data);
        });
    }

    else if (req.method === "POST" && req.url === "/login")
        { let body = "";

        req.on("data", chunk =>
            { body += chunk.toString();
        });

        req.on("end", () => {
            const { username, password } = querystring.parse(body);

            const sql = "SELECT * FROM users WHERE username = ? AND password = ? LIMIT 1";

            db.query(sql, [username, password], (err, results) => {
                if (err) throw err;

                res.writeHead(200, { "Content-Type": "text/html" });

                if (results.length > 0) {
                    res.end(`<h2 style="color:green;text-align:center;">Login Successful! Welcome
${username}</h2>`);
                } else {
                    res.end(`<h2 style="color:red;text-align:center;">Invalid Username or Password</h2>`);
                }
            });
        });
    }

    else {
        res.writeHead(404, { "Content-Type": "text/plain" });
        res.end("404 Not Found");
    }
});

server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

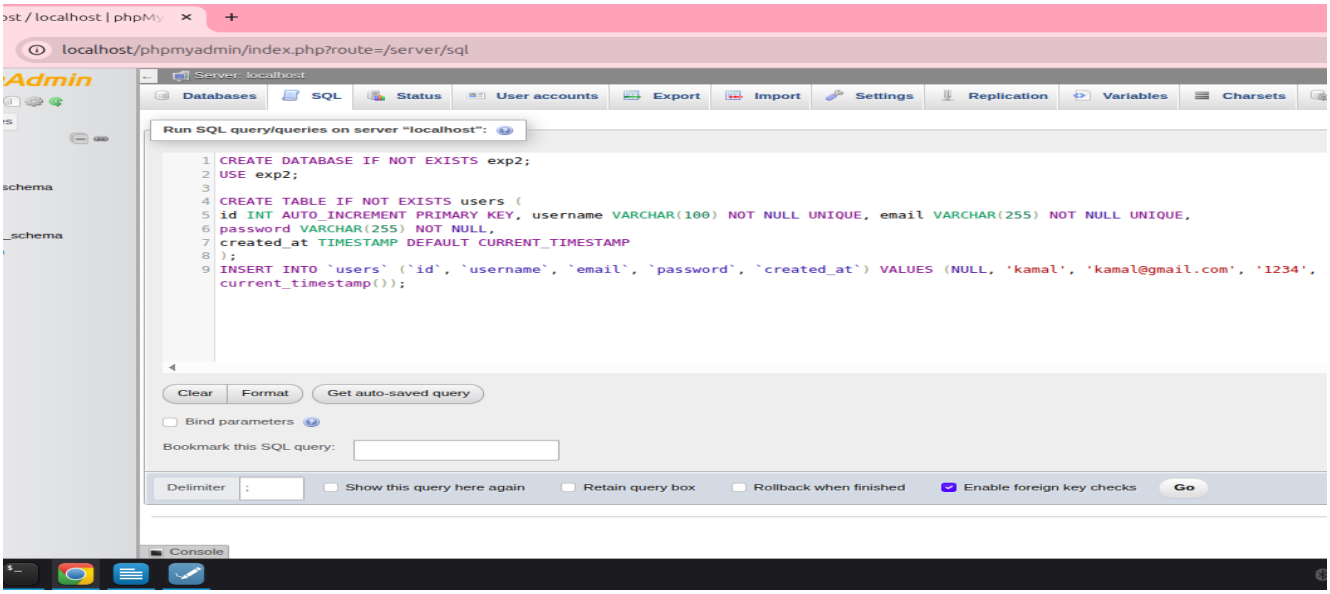
login.html :

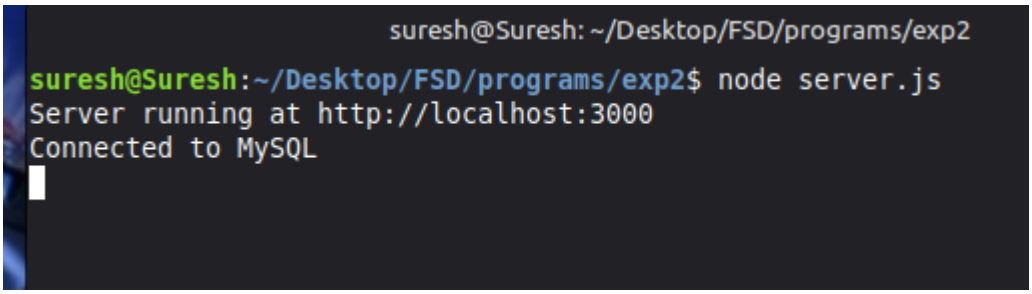
```
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Simple Login</title>
    <link rel="stylesheet" href="styles.css">
```

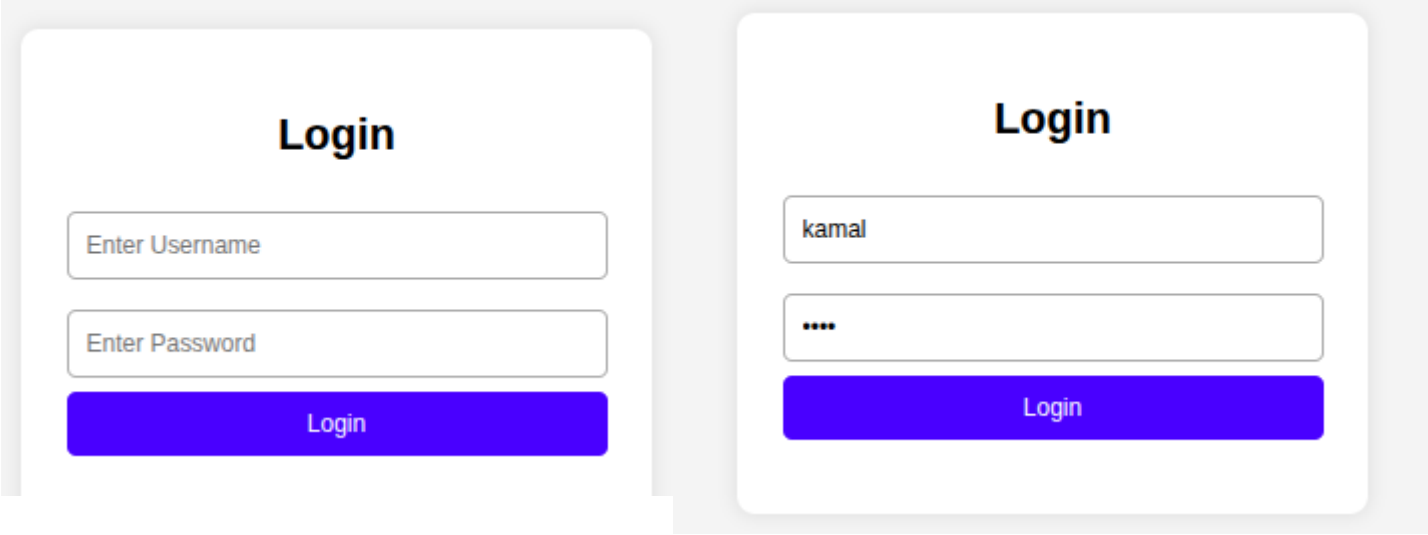
```
</head>
<body>
<div class="container">
    <h2>Login</h2>
    <form method="POST" action="/login">
        <input type="text" name="username" placeholder="Enter Username" required>
        <input type="password" name="password" placeholder="Enter Password" required>
        <button type="submit">Login</button>
    </form>

    <p id="message"></p>
</div>
</body>
</html>
```

styles.css :

```
body {
    font-family: Arial, sans-serif;
    background: #f4f4f4;
    display: flex;
    height: 100vh;
    align-items: center;
    justify-content: center;
}
.container
    { background:
    white; padding:
    25px; border-radius:
    10px; width: 300px;
    box-shadow: 0 0 10px rgba(0,0,0,0.1);
}
h2 {
    text-align: center;
}
input {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    border-radius: 5px;
    border: 1px solid #aaa;
}
button {
    width: 100%;
    padding: 10px;
    background: #007bff;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}
button:hover
    { background:
    #0056b3;
}
```



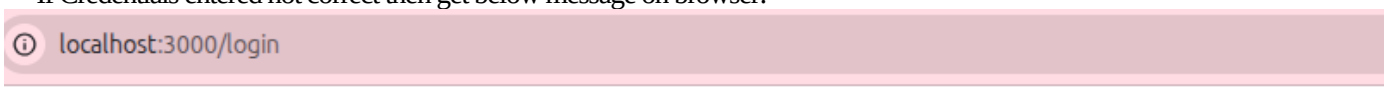




If Credentials entered correct then get below message on browser:

Login Successful! Welcome kamal

If Credentials entered not correct then get below message on browser:



Invalid Username or Password

EXPERIMENT – 3

Write a Node JS program to perform read, write and other operations on a file.

AIM: To write a Node.js program that performs **write**, **read**, **append**, **rename**, and **delete** operations on a file using the built-in `fs` module.

OBJECTIVE:

- To understand the use of Node.js **File System (fs)** module.
- To create, read, update, rename, and delete files programmatically.
- To learn asynchronous callback-based file handling.

THEORY:

Node.js provides the **fs (File System)** module to work with files.
It allows both **synchronous** and **asynchronous** operations such as:

- `writeFile()` → Create/write file
- `readFile()` → Read file
- `appendFile()` → Add data to existing file
- `rename()` → Change file name
- `unlink()` → Delete a file

In this experiment, we use **asynchronous** functions because they do not block the main thread.

PROGRAM:
file_operations.js

```
const fs = require("fs");

// 1. Write data to a file
fs.writeFile("sample.txt", "This is the initial content.\n", (err) => {
    if (err) throw err;
    console.log("File created and data written!");

    // 2. Read the file
    fs.readFile("sample.txt", "utf8", (err, data) =>
        { if (err) throw err;
        console.log("File Content:");
        console.log(data);

        // 3. Append data to the file
        fs.appendFile("sample.txt", "This line is appended.\n", (err) => {
            if (err) throw err;
            console.log("Data appended to file!");

            // 4. Rename the file
            fs.rename("sample.txt", "new_sample.txt", (err) =>
                { if (err) throw err;
                console.log("File renamed to new_sample.txt");
```

```
            // 5. Delete the file
            fs.unlink("new_sample.txt", (err) => {
                if (err) throw err;
                console.log("File deleted successfully!");
            });
        });
    });
  });
});
```

Run Procedure (Output):

```
suresh@Suresh:~/Desktop/FSD/programs$ cd exp3
suresh@Suresh:~/Desktop/FSD/programs/exp3$ node file_operations.js
File created and data written!
File Content:
This is the initial content.

Data appended to file!
This is the initial content.
This line is appended.

File renamed to new_sample.txt
File deleted successfully!
```

EXPERIMENT – 4

Write a Node JS program to read form data from query string and generate response using NodeJS

AIM

To write a Node.js program that reads form data sent through a **query string** and generates an appropriate response.

OBJECTIVE

- To understand how Node.js handles HTTP GET requests.
- To extract parameters from a URL query string.
- To generate dynamic output based on user input.

THEORY

In HTTP GET requests, form data is sent in the URL after a `?` symbol in key–value pairs.
Example:

http://localhost:3000/?name=John&age=20

Node.js provides the built-in **url** module to parse query strings:

- `url.parse(req.url, true).query` gives an object with form data.

PROGRAM:

server.js

```
const http = require("http");
const url = require("url");

const server = http.createServer((req, res) => {

    // Parse URL and get query string data
    const q = url.parse(req.url, true).query;

    // If no data, show form
    if (!q.name && !q.age) {
        res.writeHead(200, { "Content-Type": "text/html" });
        res.end(`
                    <center><div>
            <h2>Enter Your Details</h2>
            <form method="GET" action="/">
                <input type="text" name="name" placeholder="Enter Name" required><br><br>
                <input type="number" name="age" placeholder="Enter Age" required><br><br>
                <button type="submit">Submit</button>
            </form></div></center>
        `);
    }
```

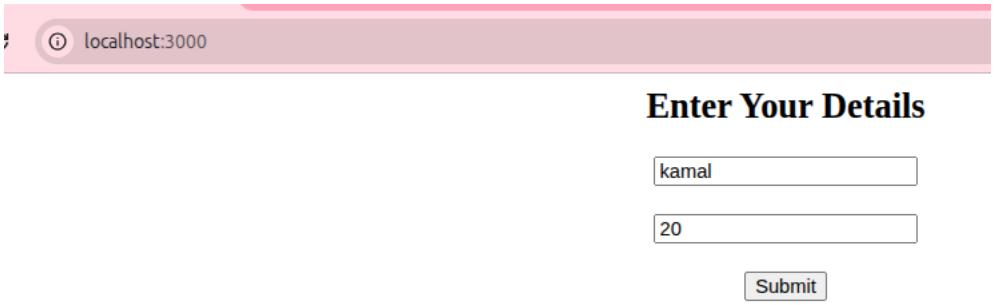
```
    else {
        // Data available → generate response
        res.writeHead(200, { "Content-Type": "text/html" });
        res.end(`
            <h2>Form Submitted Successfully!</h2>
            <p><b>Name:</b> ${q.name}</p>
            <p><b>Age:</b> ${q.age}</p>
        `);
    }
});

// Start server
server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

OUTPUT

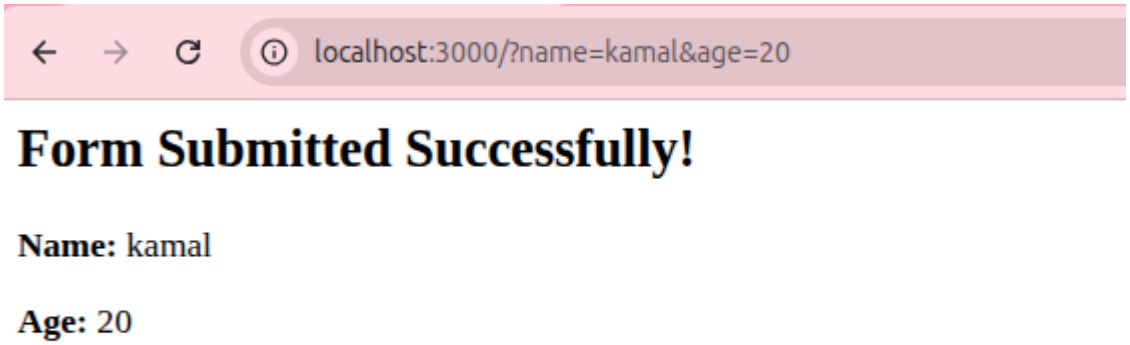
```
suresh@Suresh:~/Desktop/FSD/programs$ cd exp4
suresh@Suresh:~/Desktop/FSD/programs/exp4$ node server.js
Server running at http://localhost:3000
```

Case 1: Initial Page

Displays form:



After submitting



EXPERIMENT – 5

Create a food delivery website where users can order food from a particular restaurant listed in the website for handling http requests and responses using NodeJS.

AIM:

To create a simple food delivery website using Node.js where restaurant names, menu items, and prices are stored in a JSON file**, and to handle HTTP requests and responses.**

OBJECTIVE

- To understand HTTP request and response handling in Node.js
- To create a basic web server using Node.js
- To display restaurant and food items
- To process user order input

THEORY

Node.js can read external files using the **fs module**: In this experiment:

- Restaurant and menu details are stored in a **JSON file**
- Node.js reads the JSON file and displays restaurants and menus
- User selects food items
- Server calculates the total bill and sends response
- `styles.css` is used for styling the pages

P**ROCEDURE:**

- Create files: `server.js`, `data.json`, `styles.css`
- Run:

C:\exp5> node server.js

PROGRAM:

data.json:

```
{
  "restaurants": [
    {
      "id": "spicyhub",
      "name": "Spicy Hub",
      "menu": {
        "Biryani": 200,
        "Paneer": 150,
        "Roti": 20
      }
```

```
    },
    {
      "id": "italiandelight",
      "name": "Italian Delight",
      "menu": {
        "Pizza": 250,
        "Pasta": 180,
        "Garlic Bread": 120
      }
    }
  ]
}
```

server.js

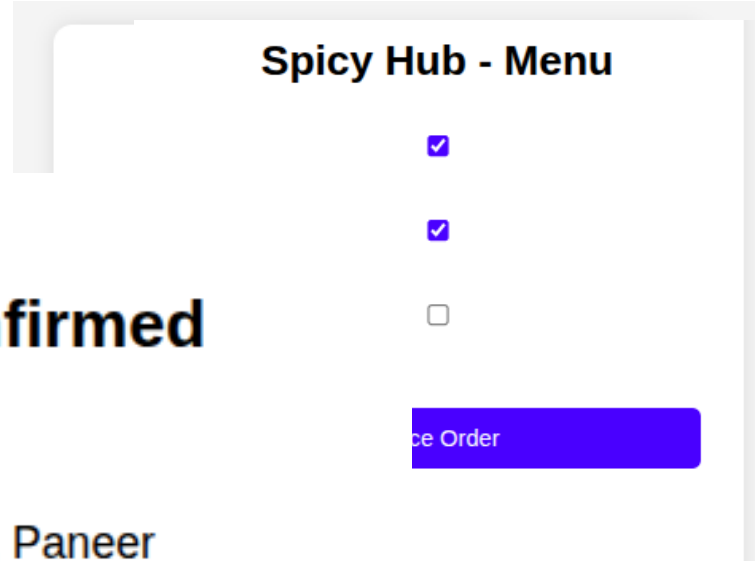
```
const http = require("http");
const fs = require("fs");
const querystring = require("querystring");

// Read JSON data
const data = JSON.parse(fs.readFileSync("data.json", "utf8"));

const server = http.createServer((req, res) => {

    // Serve CSS
    if (req.url === "/styles.css")
        { fs.readFile("styles.css", (err, css) => {
            res.writeHead(200, { "Content-Type": "text/css" });
            res.end(css);
        });
    }

    // Home Page - Restaurant List
    else if (req.url === "/" && req.method === "GET")
        { let restHtml = "";
        data.restaurants.forEach(r => {
            restHtml += `<a href="/menu?rest=${r.id}" class="btn">${r.name}</a><br><br>`;
        });

        res.writeHead(200, { "Content-Type": "text/html" });
        res.end(`
            <html>
            <head>
                <title>Food Delivery</title>
                <link rel="stylesheet" href="styles.css">
            </head>
            <body>
                <div class="container">
                    <h2>Food Delivery Website</h2>
                    <h3>Select Restaurant</h3>
                    ${restHtml}
                </div>
            </body>
            </html>
        `);
```

```
}

// Menu Page
else if (req.url.startsWith("/menu") && req.method === "GET")
    { const query = new URL(req.url, "http://localhost").searchParams;
    const restId = query.get("rest");
    const restaurant = data.restaurants.find(r => r.id === restId);

    if (!restaurant) {
        res.writeHead(404, { "Content-Type": "text/html" });
        res.end("<h3>Restaurant not found</h3>");
        return;
    }

    let menuHtml = "";
    for (let item in restaurant.menu)
        { menuHtml += `
            <input type="checkbox" name="food" value="${item}">
            ${item} (Rs. ${restaurant.menu[item]})<br>
        `;
    }

    res.writeHead(200, { "Content-Type": "text/html" });
    res.end(`
        <html>
        <head>
            <title>${restaurant.name}</title>
            <link rel="stylesheet" href="styles.css">
        </head>
        <body>
            <div class="container">
                <h2>${restaurant.name} - Menu</h2>
                <form method="POST" action="/order?rest=${restaurant.id}">
                    ${menuHtml}<br>
                    <button type="submit" class="btn">Place Order</button>
                </form>
                <br>
                <a href="/" class="link">Back</a>
            </div>
        </body>
        </html>
    `);
}

// Order Confirmation Page
else if (req.url.startsWith("/order") && req.method === "POST")
    { const query = new URL(req.url, "http://localhost").searchParams;
    const restId = query.get("rest");
    const restaurant = data.restaurants.find(r => r.id === restId);

    let body = "";
    req.on("data", chunk => body += chunk.toString());

    req.on("end", () => {
```

```
            let formData = querystring.parse(body);
            let items = formData.food || [];

            if (!Array.isArray(items)) items = [items];

            let total = 0;
            items.forEach(item => {
                total += restaurant.menu[item];
            });

            res.writeHead(200, { "Content-Type": "text/html" });
            res.end(`
                <html>
                <head>
                    <title>Order Confirmation</title>
                    <link rel="stylesheet" href="styles.css">
                </head>
                <body>
                    <div class="container">
                        <h2>Order Confirmed</h2>
                        <p><b>Restaurant:</b> ${restaurant.name}</p>
                        <p><b>Items Ordered:</b> ${items.join(", ")}</p>
                        <p><b>Total Amount to Pay:</b> ${total}</p>
                        <p class="success">Thank you for your order!</p>
                        <a href="/" class="btn">Home</a>
                    </div>
                </body>
                </html>
            `);
        });
    }

    // Invalid URL
    else {
        res.writeHead(404, { "Content-Type": "text/html" });
        res.end("<h3>404 - Page Not Found</h3>");
    }
});

// Start Server
server.listen(3000, () => {
    console.log("Server running at http://localhost:3000");
});
```

OutPut:

```
suresh@Suresh:~/Desktop/FSD/programs$ cd exp5
suresh@Suresh:~/Desktop/FSD/programs/exp5$ node server.js
Server running at http://localhost:3000
```



Order Confirmed

Restaurant: Spicy Hub

Items Ordered: Biryani, Paneer

Total Amount to Pay: 350

Thank you for your order!

Home

EXPERIMENT – 6

Implement a program with basic commands on databases and collections using MongoDB.

AIM:

To implement a Node.js program that performs **basic MongoDB commands** such as creating a database, creating a collection, inserting documents, reading documents, updating documents, and deleting documents.

OBJECTIVE

- To understand MongoDB database and collections
- To perform CRUD operations using MongoDB
- To connect Node.js with MongoDB
- To execute basic MongoDB commands programmatically

THEORY

MongoDB is a **NoSQL document-oriented database** that stores data in JSON-like documents.
In MongoDB:

- A **database** contains **collections**
- A **collection** contains **documents**
- CRUD operations include:
  - **Create** → `insertOne()`
  - **Read** → `find()`
  - **Update** → `updateOne()`
  - **Delete** → `deleteOne()`

Node.js connects to MongoDB using the official **mongodb** driver.

PROCEDURE

1. Start MongoDB server
2. Create a Node.js project folder
3. Install MongoDB driver
4. Write Node.js program to perform CRUD operations
5. Run the program and observe output

PROGRAM:

- **File name: mongo_basic.js**

```
// Import MongoClient from mongodb package
const { MongoClient } = require("mongodb");

// MongoDB connection URL
const url = "mongodb://localhost:27017";

// Create MongoClient instance
const client = new MongoClient(url);

// Database and collection names
```

```
const dbName = "collegeDB";
const collectionName = "students";

// Async function to perform MongoDB operations
async function run() {
  try {
    // Connect to MongoDB
    await client.connect();
    console.log("Connected to MongoDB");

    // Select database
    const db = client.db(dbName);

    // Select collection
    const collection = db.collection(collectionName);

    // INSERT document
    await collection.insertOne({
      rollno: 101,
      name: "Ravi",
      branch: "CSE",
      marks: 85
    });
    console.log("Document Inserted");

    // READ documents
    const students = await collection.find({}).toArray();
    console.log("Documents in Collection:");
    console.log(students);

    // UPDATE document
    await collection.updateOne(
      { rollno: 101 },
      { $set: { marks: 90 } }
    );
    console.log("Document Updated");

    // DELETE document
    await collection.deleteOne({ rollno: 101 });
    console.log("Document Deleted");

  } catch (error) {
    console.error(error);
  } finally {
    // Close MongoDB connection
    await client.close();
    console.log("Connection Closed");
  }
}

// Call function
run();
```

Execution:

OUTPUT:

1. Create a exp6 folder.
2. Change to exp6 folder
3. Check mongodb installed or not using: (if it not installed download and install from

https://www.mongodb.com/try/download/community

C:\exp6> mongod -version

```
suresh@Suresh:~/Desktop/FSD/programs/exp6$ mongod -version
db version v7.0.28
```

4. 4. Type the below command to Install MongoDB driver using npm in current Directory

C:\exp6> npm install mongodb

```
suresh@Suresh:~/Desktop/FSD/programs/exp6$ npm install mongodb

added 12 packages in 6s
suresh@Suresh:~/Desktop/FSD/programs/exp6$
```

5. Run the mongo_basic.js using node.

```
suresh@Suresh:~/Desktop/FSD/programs/exp6$ node mongo_basic.js
Connected to MongoDB
Document Inserted
Documents in Collection:
[
  {
    _id: new ObjectId('69a6edebfc90a6b9ba89070d'),
    rollno: 101,
    name: 'Ravi',
    branch: 'CSE',
    marks: 85
  }
]
Document Updated
Document Deleted
Connection Closed
```

EXPERIMENT – 7

Title : Implement CRUD operations on the given dataset using MongoDB.

Aim

To implement Create, Read, Update, and Delete (CRUD) operations on a given dataset using MongoDB and Node.js.

Objectives

- To understand CRUD operations in MongoDB
- To store and manipulate data in collections
- To connect MongoDB with Node.js
- To perform insert, read, update, and delete operations

Theory

MongoDB is a NoSQL database that stores data in **document format (JSON-like)**.
CRUD operations are the basic database operations:

- **Create** → `insertOne()` / `insertMany()`
- **Read** → `find()`
- **Update** → `updateOne()`
- **Delete** → `deleteOne()`

Program :

File Name: mongo_crud.js

```
// Import MongoClient
const { MongoClient } = require("mongodb");
// MongoDB URL
const url = "mongodb://localhost:27017";
// Create client
const client = new MongoClient(url);
// Database and collection names
const dbName = "collegeDB";
const collectionName = "cjitsstudents";
async function run() {
  try {
    // Connect to MongoDB
    await client.connect();
```

```
    console.log("Connected to MongoDB");
    // Access database and collection
    const db = client.db(dbName);
    const collection = db.collection(collectionName);
    // ---------------- CREATE ----------------
    await collection.insertMany([
      { rollno: 101, name: "Ravi", branch: "CSE", marks: 85 },
      { rollno: 102, name: "Sita", branch: "ECE", marks: 78 },
      { rollno: 103, name: "Kiran", branch: "CSE", marks: 92 }
    ]);
    console.log("Documents Inserted");
    // ---------------- READ ----------------
    const students = await collection.find({}).toArray();
    console.log("All Students:");
    console.log(students);
    // ---------------- UPDATE ----------------
    await collection.updateOne(
      { rollno: 102 },
      { $set: { marks: 82 } }
    );
  // ---------------- After Update - READ ----------------
    const updstudents = await collection.find({}).toArray();
    console.log("All Students:");
    console.log(updstudents);
    console.log("Document Updated");
    // ---------------- DELETE ----------------
    await collection.deleteOne({ rollno: 101 });
// ---------------- After Delete - READ ----------------
    const delstudents = await collection.find({}).toArray();
    console.log("All Students:");
    console.log(delstudents);
     console.log("Document Deleted");
  } catch (err) {
    console.error(err);
```

```
  } finally {
    // Close connection
    await client.close();
    console.log("Connection Closed");
  }
}
// Execute function
run();
```

Output :

C:\>md exp7

C:\>cd exp7

Node.js interacts with MongoDB using the official mongodb driver. For this install mongodb module.

C:\exp7>npm install mongodb

```
suresh@Suresh:~/Desktop/FSD/programs/exp7$ npm install mongodb

added 12 packages, and audited 13 packages in 4s

found 0 vulnerabilities
```

C:\exp7>node mongo_crud.js

```
suresh@Suresh:~/Desktop/FSD/programs/exp7$ node mongo_crud.js
Connected to MongoDB
Documents Inserted
All Students:
[
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad40e'),
    rollno: 101,
    name: 'Ravi',
    branch: 'CSE',
    marks: 85
  },
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad40f'),
    rollno: 102,
    name: 'Sita',
    branch: 'ECE',
    marks: 78
  },
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad410'),
    rollno: 103,
    name: 'Kiran',
    branch: 'CSE',
    marks: 92
  }
]
```

```
All Students:
[
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad40e'),
    rollno: 101,
    name: 'Ravi',
    branch: 'CSE',
    marks: 85
  },
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad40f'),
    rollno: 102,
    name: 'Sita',
    branch: 'ECE',
    marks: 82
  },
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad410'),
    rollno: 103,
    name: 'Kiran',
    branch: 'CSE',
    marks: 92
  }
]
Document Updated
```

```
All Students:
[
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad40f'),
    rollno: 102,
    name: 'Sita',
    branch: 'ECE',
    marks: 82
  },
  {
    _id: new ObjectId('69a6f076a59e62d7f91ad410'),
    rollno: 103,
    name: 'Kiran',
    branch: 'CSE',
    marks: 92
  }
]
Document Deleted
Connection Closed
```

EXPERIMENT – 8

Title : Perform Count, Limit, Sort, and Skip operations on the given collections using MongoDB.

Aim : To perform Count, Limit, Sort, and Skip operations on a MongoDB collection using Node.js.

Objectives

- To understand advanced read operations in MongoDB
- To retrieve limited and sorted records
- To implement pagination using skip and limit
- To connect Node.js with MongoDB

Theory

MongoDB provides several query methods to control the number and order of documents returned.

- **Count** → Counts number of documents
- **Limit** → Restricts number of documents returned
- **Sort** → Arranges documents in ascending or descending order
- **Skip** → Skips specified number of documents

These operations are commonly used in pagination and report generation.

Dataset Used (csestudents collection)

rollno	name	branch	marks
501	Pavan	CSE	15
502	Sita	CSE	58
503	kamal	CSE	22

Program

File Name: count_limit_sort_skip.js

```
// Import MongoClient

const { MongoClient } = require("mongodb");

// MongoDB URL
```

```
const url = "mongodb://localhost:27017";

// Create MongoClient instance

const client = new MongoClient(url);

// Database and collection names

const dbName = "collegeDB";

const collectionName = "csestudents";

async function run() {

  try {

    // Connect to MongoDB

    await client.connect();

    console.log("Connected to MongoDB");

    const db = client.db(dbName);

    const collection = db.collection(collectionName);

  // ---------------- CREATE ----------------

    await collection.insertMany([

      { rollno: 501, name: "Pavan", branch: "CSE", marks: 15 },

      { rollno: 502, name: "Sita", branch: "CSE", marks: 58 },

      { rollno: 503, name: "Kamal", branch: "CSE", marks: 22 }

    ]);

    console.log("Documents Inserted");

    // ---------------- COUNT ----------------

    const count = await collection.countDocuments();

    console.log("Total Number of Students:", count);

    // ---------------- SORT ----------------

    const sortedStudents = await collection

      .find({})
```

```
    .sort({ marks: -1 })   // Descending order
    .toArray();
  console.log("Students Sorted by Marks (Descending):");
  console.log(sortedStudents);
  // ---------------- LIMIT ----------------
  const limitedStudents = await collection
    .find({})
    .limit(3)
    .toArray();
  console.log("First 3 Students:");
  console.log(limitedStudents);
  // ---------------- SKIP + LIMIT ----------------
  const skippedStudents = await collection
    .find({})
    .skip(2)
    .limit(2)
    .toArray();
  console.log("Students After Skipping 2 Records:");
  console.log(skippedStudents);
} catch (error) {
  console.error(error);
} finally {
  // Close connection
  await client.close();
  console.log("Connection Closed");
}
```

}

// Execute function

run();

Output :

C:\>md exp8

C:\>cd exp8

Node.js interacts with MongoDB using the official mongodb driver. For this install mongodb module.

C:\exp8>npm install mongodb

```
suresh@Suresh:~/Desktop/FSD/programs/exp8$ npm install mongodb

added 12 packages in 4s
```

C:\exp7>node count_limit_sort_skip.js

```
suresh@Suresh:~/Desktop/FSD/programs/exp8$ node count_limit_sort_skip.js
Connected to MongoDB
Documents Inserted
Total Number of Students: 3
Students Sorted by Marks (Descending):
[
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fce'),
    rollno: 502,
    name: 'Sita',
    branch: 'CSE',
    marks: 58
  },
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fcf'),
    rollno: 503,
    name: 'Kamal',
    branch: 'CSE',
    marks: 22
  },
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fcd'),
    rollno: 501,
    name: 'Pavan',
    branch: 'CSE',
    marks: 15
  }
]
```

```
First 3 Students:
[
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fcd'),
    rollno: 501,
    name: 'Pavan',
    branch: 'CSE',
    marks: 15
  },
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fce'),
    rollno: 502,
    name: 'Sita',
    branch: 'CSE',
    marks: 58
  },
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fcf'),
    rollno: 503,
    name: 'Kamal',
    branch: 'CSE',
    marks: 22
  }
]
Students After Skipping 2 Records:
[
  {
    _id: new ObjectId('69a6f44f44b2bf2962218fcf'),
    rollno: 503,
    name: 'Kamal',
    branch: 'CSE',
    marks: 22
  }
]
Connection Closed
```

EXPERIMENT – 9

Title: Develop an AngularJS form to apply CSS and Events.

AIM: To develop an AngularJS form and apply CSS styling and event handling using AngularJS directives.

Objectives

- To design a form using AngularJS
- To apply CSS styles dynamically
- To handle user events like button click
- To understand AngularJS directives.

Theory

AngularJS is a JavaScript framework used to create dynamic web applications.
It supports:

- **Two-way data binding**
- **Directives** such as `ng-app`, `ng-model`, `ng-click`
- **Event handling** without writing JavaScript separately

CSS is used to improve the appearance of HTML elements, while AngularJS events respond to user actions.

Procedure

1. Create an HTML file
2. Include AngularJS library
3. Define AngularJS application and controller
4. Design a form with input fields
5. Apply CSS styles
6. Handle events using AngularJS directives

Program

File Name: `index.html`

```
<!DOCTYPE html>
<html lang="en" ng-app="myApp">
<head>
  <meta charset="UTF-8">
  <title>AngularJS Form with CSS and Events</title>
```

```
  <!-- AngularJS Library -->
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>

  <!-- CSS Styling -->
  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
    }
    .container {
      width: 350px;
      margin: auto;
      margin-top: 80px;
      padding: 20px;
      background: white;
      border-radius: 10px;
      box-shadow: 0px 0px 10px gray;
    }
    h2 {
      text-align: center;
      color: #007bff;
    }
    input {
      width: 100%;
      padding: 10px;
      margin: 8px 0;
      border-radius: 5px;
      border: 1px solid #aaa;
    }
    button {
      width: 100%;
      padding: 10px;
      background-color: green;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }
    button:hover {
      background-color: darkgreen;
    }
    .result {
      margin-top: 15px;
      color: blue;
      text-align: center;
    }
  </style>
</head>
```

```
<body ng-controller="myCtrl">

  <div class="container">
    <h2>User Form</h2>

    <label>Name</label>
    <input type="text" ng-model="name" placeholder="Enter Name">

    <label>Email</label>
    <input type="email" ng-model="email" placeholder="Enter Email">

    <button ng-click="submit()">Submit</button>

    <div class="result" ng-if="submitted">
      <p>Name: {{name}}</p>
      <p>Email: {{email}}</p>
    </div>
  </div>

  <script>
    var app = angular.module("myApp", []);
    app.controller("myCtrl", function($scope) {
      $scope.submitted = false;

      $scope.submit = function() {
        $scope.submitted = true;
      };
    });
  </script>

</body>
</html>
```

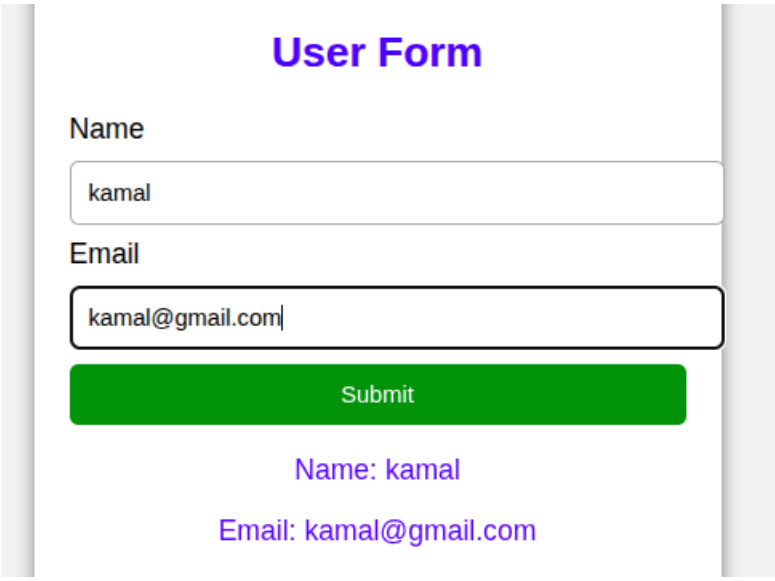
Output:

1. Create exp9 folder and write program code here.

C:\>md exp9

2. Type the `index.html code in this folder.`

2.Double click on the index.html, it opens in the browser



- Displays a styled form
- Accepts user input
- On button click, entered details are displayed dynamically

EXPERIMENT – 10

Title: Develop a Job Registration form and validate it using AngularJS.

AIM: To develop a Job Registration form using AngularJS and perform client-side form validation.

Objectives

- To design a job registration form
- To apply AngularJS form validation
- To understand AngularJS validation directives
- To prevent submission of invalid data

Theory

AngularJS provides built-in form validation using directives such as:

- `ng-model`
- `ng-required`
- `ng-pattern`
- `$valid`, `$invalid`, `$touched`

AngularJS automatically tracks form and input states and displays error messages dynamically without page reload.

Procedure

1. Create an HTML file
2. Include AngularJS library
3. Define AngularJS module and controller
4. Design Job Registration form
5. Apply validation rules
6. Display error messages
7. Run the application in browser

Program

File Name: `job_registration.html`

```
<!DOCTYPE html>
<html lang="en" ng-app="jobApp">
<head>
  <meta charset="UTF-8">
  <title>Job Registration Form</title>

  <!-- AngularJS Library -->
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>

  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
    }
    .container {
      width: 400px;
      margin: auto;
      margin-top: 60px;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0px 0px 10px gray;
    }
    h2 {
      text-align: center;
      color: #007bff;
    }
    input, select {
      width: 100%;
      padding: 8px;
      margin: 6px 0;
    }
    button {
      width: 100%;
      padding: 10px;
      background: green;
      color: white;
      border: none;
      border-radius: 5px;
    }
    .error {
      color: red;
```

```
      font-size: 13px;
    }
    .success {
      color: green;
      text-align: center;
      margin-top: 10px;
    }
  </style>
</head>

<body ng-controller="jobCtrl">

  <div class="container">
    <h2>Job Registration</h2>

    <form name="jobForm" novalidate>

      <!-- Name -->
      <label>Full Name</label>
      <input type="text" name="name"
             ng-model="user.name"
             ng-required="true">
      <span class="error"
            ng-show="jobForm.name.$touched && jobForm.name.$invalid">
            Name is required
      </span>

      <!-- Email -->
      <label>Email</label>
      <input type="email" name="email"
             ng-model="user.email"
             ng-required="true">
      <span class="error"
            ng-show="jobForm.email.$touched && jobForm.email.$invalid">
            Enter valid email
      </span>

      <!-- Mobile -->
      <label>Mobile Number</label>
      <input type="text" name="mobile"
             ng-model="user.mobile"
             ng-pattern="/^[0-9]{10}$/"
             ng-required="true">
      <span class="error"
            ng-show="jobForm.mobile.$touched && jobForm.mobile.$invalid">
            Enter 10 digit mobile number
```

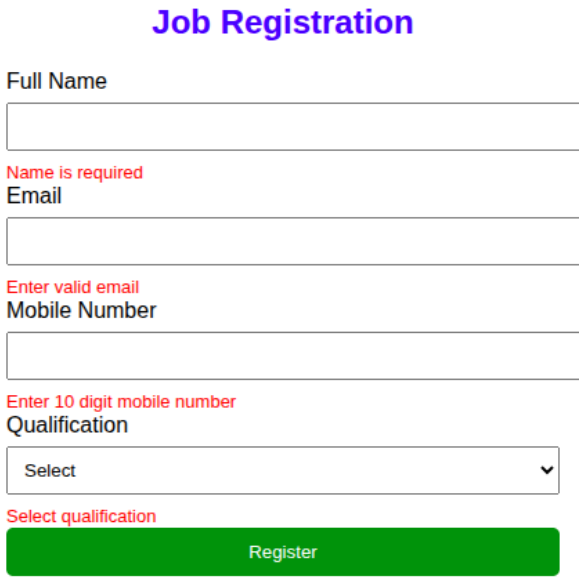
```
    </span>

    <!-- Qualification -->
    <label>Qualification</label>
    <select name="qualification"
            ng-model="user.qualification"
            ng-required="true">
      <option value="">Select</option>
      <option>B.Tech</option>
      <option>M.Tech</option>
      <option>MCA</option>
      <option>B.Sc</option>
    </select>
    <span class="error"
          ng-show="jobForm.qualification.$touched && jobForm.qualification.$invalid">
          Select qualification
    </span>

    <!-- Submit -->
    <button type="submit"
            ng-click="register()"
            ng-disabled="jobForm.$invalid">
            Register
    </button>
  </form>
  <div class="success" ng-if="success">
    Registration Successful!
  </div>
</div>

<script>
  var app = angular.module("jobApp", []);
  app.controller("jobCtrl", function($scope) {
    $scope.success = false;

    $scope.register = function() {
      if ($scope.jobForm.$valid) {
        $scope.success = true;
      }
    };
  });
</script>

</body>
</html>
```

Output:

1. Double click on the `job_registration.html` file, it open in the browser.

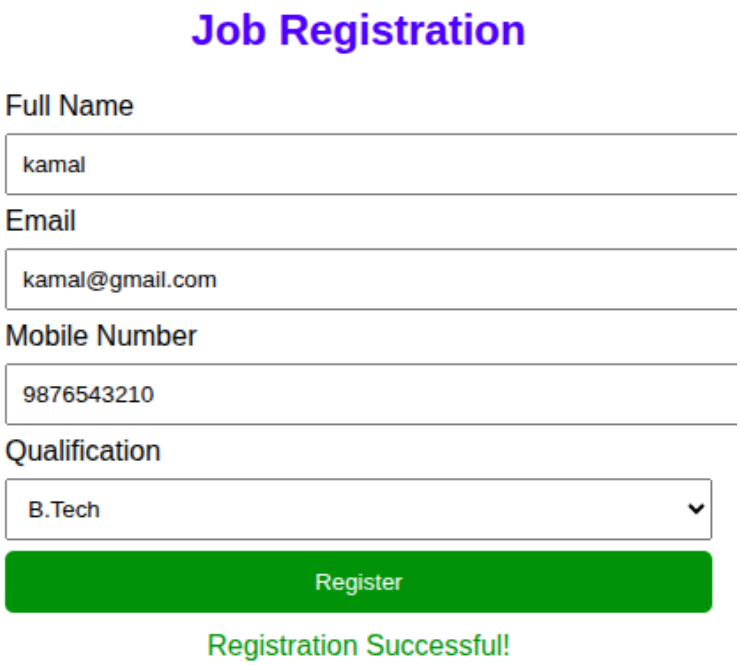
- Displays Job Registration Form

2. Shows error messages for invalid inputs



- Submit button enabled only when form is valid

3. Displays **Registration Successful** message on valid submission.



EXPERIMENT – 11

Title: Write an AngularJS application to access JSON file data of an employee from a server using `$http` service.

AIM: To develop an AngularJS application that retrieves employee data from a JSON file using the `$http` service and displays it on a web page.

Objectives

- To understand `$http` service in AngularJS
- To fetch data from a JSON file (server)
- To display dynamic data using AngularJS
- To understand promises (`then`, `catch`)

Theory:

AngularJS provides the `$http` service to communicate with servers using HTTP methods like **GET, POST, PUT, DELETE**.

In this experiment:

- Employee details are stored in a **JSON file**
- `$http.get()` fetches data from the server
- Retrieved data is bound to the view using AngularJS expressions

Procedure

1. Create a JSON file with employee data
2. Create an HTML file
3. Include AngularJS library
4. Define AngularJS module and controller
5. Use `$http.get()` to fetch JSON data
6. Display employee details in a table

Program

File 1: `employees.json`

```
[
  {
    "empId": 101,
```

```
    "name": "Ravi",
    "designation": "Software Engineer",
    "salary": 45000
  },
  {
    "empId": 102,
    "name": "Sita",
    "designation": "Tester",
    "salary": 38000
  },
  {
    "empId": 103,
    "name": "Kiran",
    "designation": "Project Manager",
    "salary": 60000
  }
]
```

File 2: `index.html`

```
<!DOCTYPE html>
<html lang="en" ng-app="empApp">
<head>
  <meta charset="UTF-8">
  <title>Employee Details</title>

  <!-- AngularJS Library -->
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>

  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
    }
    .container {
      width: 600px;
      margin: auto;
      margin-top: 60px;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0px 0px 10px gray;
    }
    h2 {
      text-align: center;
      color: #007bff;
```

```
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 15px;
    }
    th, td {
      padding: 10px;
      border: 1px solid #aaa;
      text-align: center;
    }
    th {
      background-color: #007bff;
      color: white;
    }
  </style>
</head>

<body ng-controller="empCtrl">

  <div class="container">
    <h2>Employee Details</h2>

    <table>
      <tr>
        <th>Employee ID</th>
        <th>Name</th>
        <th>Designation</th>
        <th>Salary</th>
      </tr>
      <tr ng-repeat="emp in employees">
        <td>{{emp.empId}}</td>
        <td>{{emp.name}}</td>
        <td>{{emp.designation}}</td>
        <td>{{emp.salary}}</td>
      </tr>
    </table>
  </div>

  <script>
    var app = angular.module("empApp", []);

    app.controller("empCtrl", function($scope, $http) {

      // Fetch JSON data using $http service
      $http.get("employees.json")
```

```
      .then(function(response) {
        $scope.employees = response.data;
      })
      .catch(function(error) {
        console.error("Error fetching data", error);
      });

  });
</script>

</body>
</html>
```

Output

- Employee details are fetched from employees.json
- Data is displayed dynamically in a table
- Page updates without reload

1. Create a exp10 folder in the C:\ Drive
2. Type index.html code, and employees.json file in the exp10 folder.
3. install http module in the current folder and start http server.

```
suresh@Suresh:~/Desktop/FSD/programs/exp10$ npm install -g http

added 1 package in 781ms
suresh@Suresh:~/Desktop/FSD/programs/exp10$ http-server
Starting up http-server, serving ./

http-server version: 14.1.1

http-server settings:
CORS: disabled
Cache: 3600 seconds
Connection Timeout: 120 seconds
Directory Listings: visible
AutoIndex: visible
Serve GZIP Files: false
Serve Brotli Files: false
Default File Extension: none

Available on:
  http://127.0.0.1:8080
  http://192.168.100.108:8080
Hit CTRL-C to stop the server
```

4. Open the browser and type http://127.0.0.1:8080 it shows the records of employees.json file

Employee Details

Employee ID	Name	Designation	Salary
101	Ravi	Software Engineer	45000
102	Sita	Tester	38000
103	Kiran	Project Manager	60000

EXPERIMENT – 12

Title: Develop a web application to manage student information using Express and AngularJS.

AIM: To develop a web application using Express (Node.js) and AngularJS to manage student information such as adding and viewing student details.

Objectives

- To understand Express framework
- To create REST API using Express
- To fetch data using AngularJS `$http` service
- To manage student information dynamically

Theory

Express.js is a lightweight Node.js framework used to create web servers and REST APIs.

AngularJS is a front-end framework used to build dynamic single-page applications.

In this application:

- Express acts as **backend server**
- AngularJS acts as **frontend**
- Student data is exchanged using **JSON format**

Procedure

1. Create project folder
2. Initialize Node.js project
3. Install Express
4. Create server using Express
5. Create AngularJS frontend
6. Perform add and view student operations

Project Structure

```
exp12/
├── server.js
└── public/
    └── index.html
```

Program

File 1: `server.js` (Express Backend)

```
const express = require("express");
const app = express();
const port = 3000;

app.use(express.json());
app.use(express.static("public"));

// In-memory student data
let students = [
  { rollno: 101, name: "Ravi", branch: "CSE" },
  { rollno: 102, name: "Sita", branch: "ECE" }
];

// READ – Get all students
app.get("/students", (req, res) => {
  res.json(students);
});

// CREATE – Add new student
app.post("/students", (req, res) => {
  students.push(req.body);
  res.json({ message: "Student Added" });
});

// UPDATE – Edit student
app.put("/students/:rollno", (req, res) => {
  const roll = parseInt(req.params.rollno);
  const index = students.findIndex(s => s.rollno === roll);

  if (index !== -1) {
    students[index] = req.body;
    res.json({ message: "Student Updated" });
  } else {
    res.status(404).json({ message: "Student Not Found" });
  }
});

// DELETE – Remove student
app.delete("/students/:rollno", (req, res) => {
  const roll = parseInt(req.params.rollno);
  students = students.filter(s => s.rollno !== roll);
  res.json({ message: "Student Deleted" });
});

// Start server
app.listen(port, () => {
  console.log(`Server running at http://localhost:${port}`);
});
```

File 2: `public/index.html` (AngularJS Frontend):

```
<!DOCTYPE html>
<html ng-app="studentApp">
<head>
  <title>Student Management</title>

  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>

  <style>
    body { font-family: Arial; background: #f2f2f2; }
    .container {
      width: 600px;
      margin: auto;
      background: white;
      padding: 20px;
      margin-top: 40px;
      border-radius: 10px;
    }
    input {
      width: 100%;
      padding: 8px;
      margin: 6px 0;
    }
    button {
      padding: 8px;
      margin: 5px;
      border: none;
      cursor: pointer;
    }
    .add { background: green; color: white; width: 100%; }
    .edit { background: orange; color: white; }
    .delete { background: red; color: white; }
    table {
      width: 100%;
      margin-top: 15px;
      border-collapse: collapse;
    }
    th, td {
      border: 1px solid #aaa;
      padding: 8px;
      text-align: center;
    }
    th {
      background: #007bff;
      color: white;
    }
  </style>
</head>

<body ng-controller="studentCtrl">

<div class="container">
  <h2 align="center">Student Management System</h2>
```

```
  <input type="number" ng-model="student.rollno" placeholder="Roll No">
  <input type="text" ng-model="student.name" placeholder="Name">
  <input type="text" ng-model="student.branch" placeholder="Branch">

  <button class="add" ng-click="saveStudent()">
    {{ editMode ? 'Update Student' : 'Add Student' }}
  </button>

  <table>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Actions</th>
    </tr>
    <tr ng-repeat="s in students">
      <td>{{s.rollno}}</td>
      <td>{{s.name}}</td>
      <td>{{s.branch}}</td>
      <td>
        <button class="edit" ng-click="editStudent(s)">Edit</button>
        <button class="delete" ng-click="deleteStudent(s.rollno)">Delete</button>
      </td>
    </tr>
  </table>
</div>

<script>
  var app = angular.module("studentApp", []);

  app.controller("studentCtrl", function($scope, $http) {

    $scope.editMode = false;

    // Load students
    function loadStudents() {
      $http.get("/students").then(function(res) {
        $scope.students = res.data;
      });
    }
    loadStudents();

    // Add or Update student
    $scope.saveStudent = function() {
      if ($scope.editMode) {
        $http.put("/students/" + $scope.student.rollno, $scope.student)
          .then(function() {
            loadStudents();
            $scope.student = {};
            $scope.editMode = false;
          });
      } else {
        $http.post("/students", $scope.student)
```

```
        .then(function() {
          loadStudents();
          $scope.student = {};
        });
      }
    };

    // Edit student
    $scope.editStudent = function(s) {
      $scope.student = angular.copy(s);
      $scope.editMode = true;
    };

    // Delete student
    $scope.deleteStudent = function(rollno) {
      $http.delete("/students/" + rollno)
        .then(function() {
          loadStudents();
        });
    };

  });
</script>

</body>
</html>
```

How to Run:

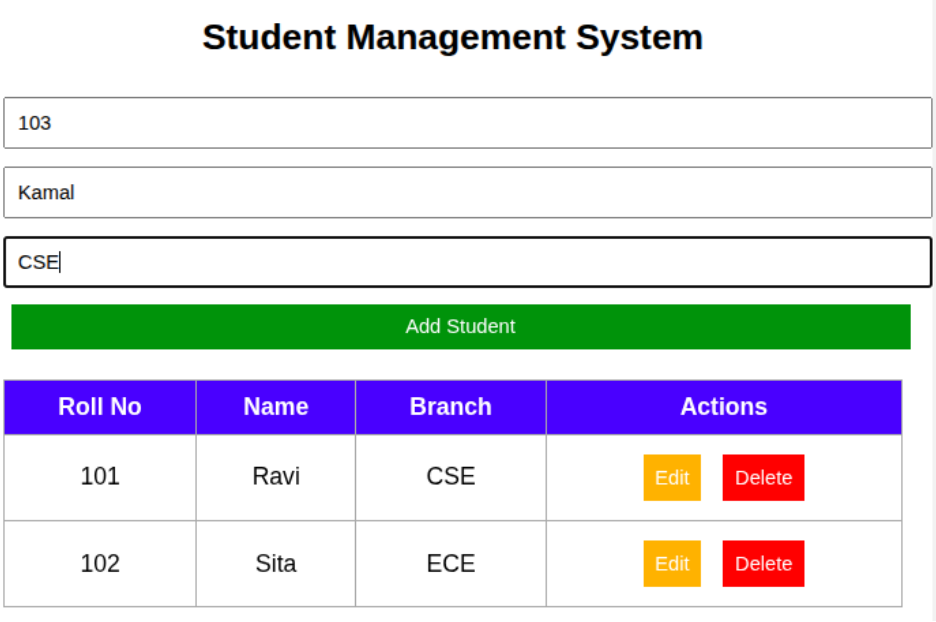
1. Create a exp12 folder and change to exp12 folder.
2. Run command: c:\exp12>npm init -y
3. Install express module using command C:\exp12>npm install express

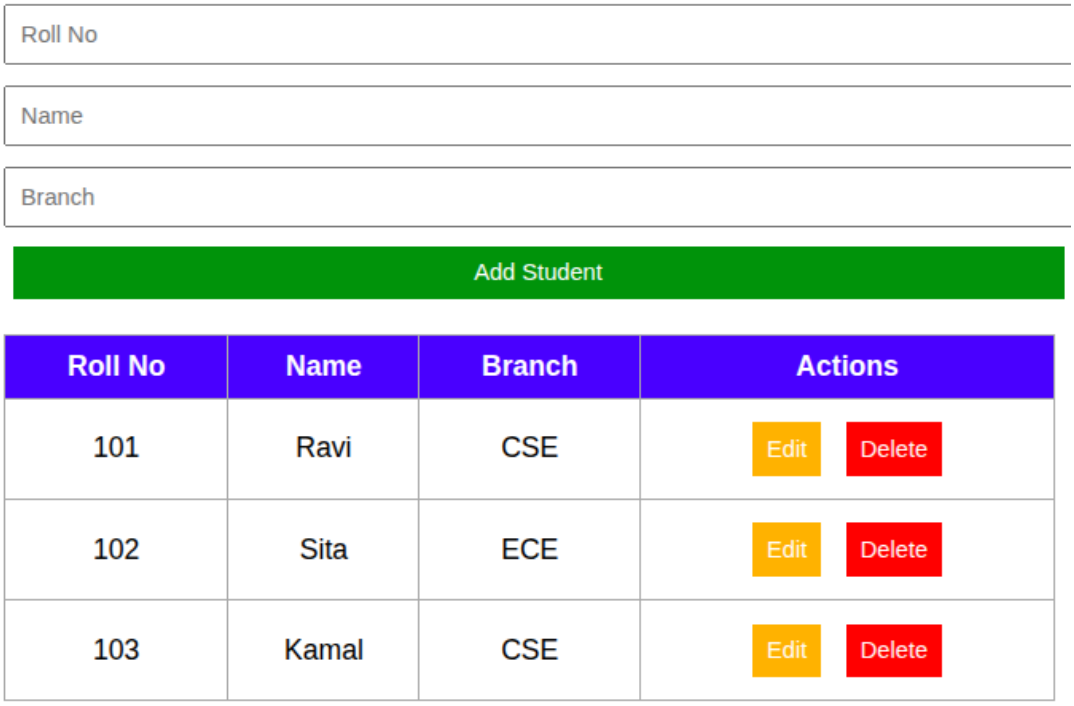
```
suresh@Suresh:~/Desktop/FSD/programs/exp12$ npm install express

up to date, audited 66 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
```

4. Run server.js file using node.

```
suresh@Suresh:~/Desktop/FSD/programs/exp12$ node server.js
Server running at http://localhost:3000
```

5. Open Browser type http://localhost:3000 at URL





EXPERIMENT – 13

Title: Write a program to create a simple calculator Application using React JS.

AIM:To develop a simple calculator application using React JS to perform basic arithmetic operations.

Objectives

- To understand React JS components
- To use `useState` hook
- To perform arithmetic operations
- To handle events in React JS

Theory

React JS is a JavaScript library used to build user interfaces using **components**.

A calculator application demonstrates:

- Component-based architecture
- State management using `useState`
- Event handling using `onClick`

Procedure

1. Create React application
2. Modify `App.js` file
3. Add calculator UI
4. Implement arithmetic logic
5. Run the application

Program

1. Create a exp13 folder , and change to exp3.

2. Type a command to create React Application :

C:\exp13>npx create-react-app calculator-app

3. Type the command following:

C:\exp13>cd calculator-app

4. Following two files App.js and App.css move to C:\exp13\calculator-app\src folder

(Overwrite with existing files)

Step 1: `App.js`

```
import React, { useState } from "react";
import "./App.css";

function App() {
  const [num1, setNum1] = useState("");
  const [num2, setNum2] = useState("");
  const [result, setResult] = useState("");

  return (
    <div className="container">
      <h2>Simple Calculator</h2>

      <input
        type="number"
        placeholder="Enter Number 1"
        value={num1}
        onChange={(e) => setNum1(e.target.value)}
      />

      <input
        type="number"
        placeholder="Enter Number 2"
        value={num2}
        onChange={(e) => setNum2(e.target.value)}
      />

      <div className="buttons">
        <button onClick={() => setResult(Number(num1) + Number(num2))}>+</button>
        <button onClick={() => setResult(Number(num1) - Number(num2))}>-</button>
        <button onClick={() => setResult(Number(num1) * Number(num2))}>*</button>
        <button onClick={() => setResult(Number(num1) / Number(num2))}>/</button>
      </div>

      <h3>Result: {result}</h3>
    </div>
  );
}

export default App;
```

Step 2: `App.css`

```
body {
  background: #f2f2f2;
  font-family: Arial;
}

.container {
  width: 300px;
  margin: auto;
  margin-top: 80px;
```

```
  background: white;
  padding: 20px;
  border-radius: 10px;
  text-align: center;
  box-shadow: 0 0 10px gray;
}

input {
  width: 100%;
  padding: 8px;
  margin: 8px 0;
}
.buttons button {
  width: 50px;
  height: 40px;
  margin: 5px;
  font-size: 18px;
  cursor: pointer;
}
```

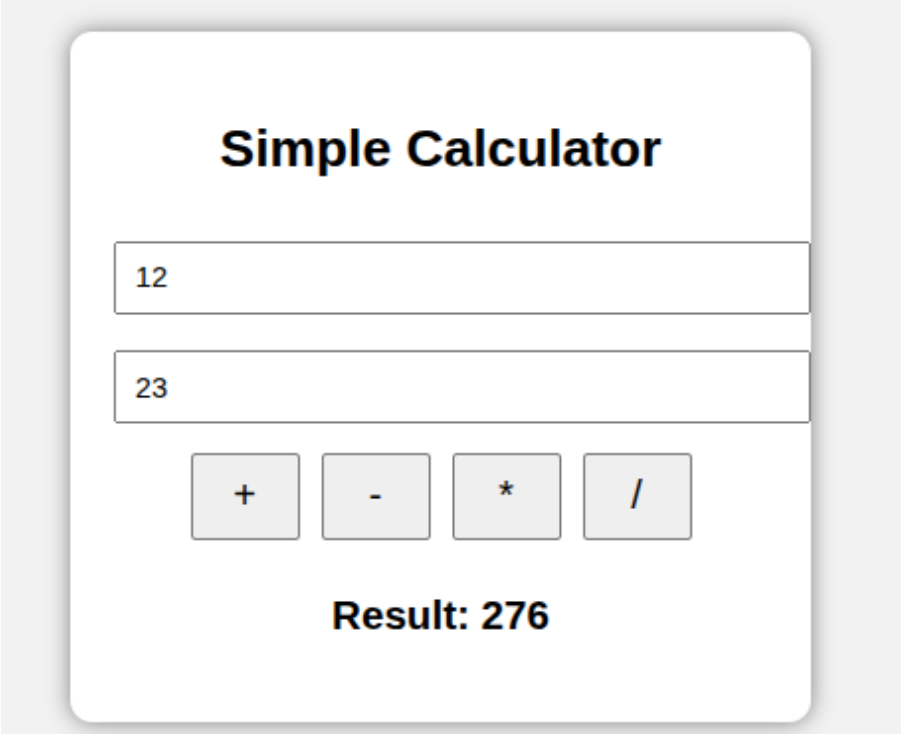
Output:

1. Type the below command to run React Application.

C:\exp13\calculator-app>npm start

```
Compiled successfully!

You can now view calculator-app in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.100.108:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

- **Displays calculator interface**



- **Performs addition, subtraction, multiplication, and division**
- **Result updates dynamically without page reload**

EXPERIMENT – 14

Title: Write a program to create a voting application using React JS

AIM:To develop a simple voting application using React JS that allows users to vote for candidates and displays vote counts dynamically.

Objectives

- To understand React JS functional components
- To use `useState` hook for state management
- To handle button click events
- To update UI dynamically without page reload

Theory

React JS is a component-based JavaScript library used to build interactive user interfaces.
In a voting application:

- Each candidate has a vote count
- Clicking a button updates the state
- React automatically re-renders the UI

Program :

1. Create a exp14 folder , and change directory to exp14.

2. Type a command to create React Application :

C:\exp14>npm create vite@latest votting-app

3. Type the command following:

C:\exp14>cd votting-app

4. Following two files App.tsx and App.css move to C:\exp13\calculator-app\src folder

(Overwrite with existing files)

Step 2: `App.tsx`

```
import React, { useState } from "react";
import "./App.css";
function App() {
  const [votesA, setVotesA] = useState(0);
```

```
  const [votesB, setVotesB] = useState(0);
  const [votesC, setVotesC] = useState(0);

  return (
    <div className="container">
      <h2>Voting Application</h2>
      <div className="card">
        <h3>Candidate A</h3>
        <p>Votes: {votesA}</p>
        <button onClick={() => setVotesA(votesA + 1)}>Vote</button>
      </div>
      <div className="card">
        <h3>Candidate B</h3>
        <p>Votes: {votesB}</p>
        <button onClick={() => setVotesB(votesB + 1)}>Vote</button>
      </div>
      <div className="card">
        <h3>Candidate C</h3>
        <p>Votes: {votesC}</p>
        <button onClick={() => setVotesC(votesC + 1)}>Vote</button>
      </div>
    </div>
  );
}
export default App;
```

Step 3: App.css

```
body {
  background-color: #f2f2f2;
  font-family: Arial;
}
.container {
  width: 400px;
```

```
  margin: auto;
  margin-top: 60px;
  background: white;
  padding: 20px;
  border-radius: 10px;
  text-align: center;
  box-shadow: 0 0 10px gray;
}
.card {
  border: 1px solid #ccc;
  margin: 10px 0;
  padding: 15px;
  border-radius: 8px;
}
button {
  padding: 8px 15px;
  background-color: green;
  color: white;
  border: none;
  cursor: pointer;
  border-radius: 5px;
}
button:hover {
  background-color: darkgreen;
}
```

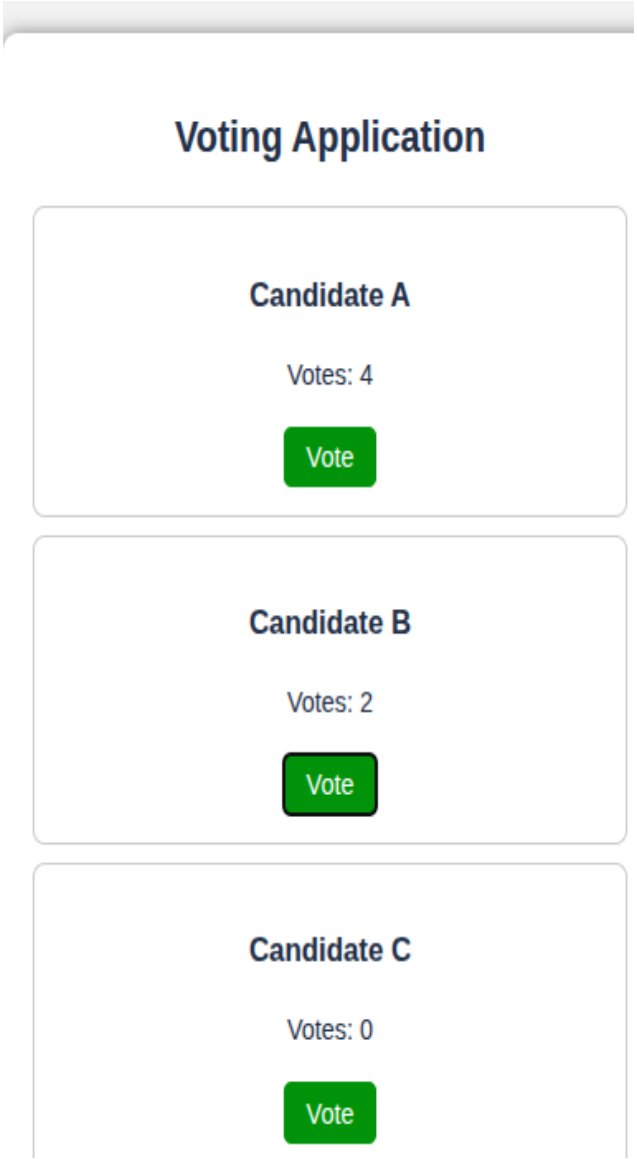
Output:

*** Type the following command to run:**

C:\exp14\votting-app> npm run dev

```
VITE v7.3.1  ready in 154 ms

➜  Local:   http://localhost:5173/
➜  Network: use --host to expose
➜  press h + enter to show help
```

*** Open the Browser and type the http://localhost:5173 at address bar**



- Displays list of candidates
- Allows user to vote
- Vote count increases dynamically